

Matrix obstructions cause multiscale disruption in collective epithelial migration by suppressing leader cell function

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ABSTRACT During disease and development, physical changes in extracellular matrix cause jamming, unjamming, and scattering in epithelial migration. However, whether disruptions in matrix topology alter collective cell migration speed and cell–cell coordination remains unclear. We microfabricated substrates with stumps of defined geometry, density, and orientation, which create obstructions for migrating epithelial cells. Here, we show that cells lose their speed and directionality when moving through densely spaced obstructions. Although leader cells are stiffer than follower cells on flat substrates, dense obstructions cause overall cell softening. Through a lattice-based model, we identify cellular protrusions, cell–cell adhesions, and leader–follower communication as key mechanisms for obstruction-sensitive collective cell migration. Our modeling predictions and experimental validations show that cells' obstruction sensitivity requires an optimal balance of cell–cell adhesions and protrusions. Both MDCK (more cohesive) and α -catenin–depleted MCF10A cells were less obstruction sensitive than wild-type MCF10A cells. Together, microscale softening, mesoscale disorder, and macroscale multicellular communication enable epithelial cell populations to sense topological obstructions encountered in challenging environments. Thus, obstruction–sensitivity could define “mechanotype” of cells that collectively migrate yet maintain intercellular communication.

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INTRODUCTION

Collective systems are capable of complex motion driven by individual motile entities that operate at varying lengths and timescales. A flock of birds can produce swirls and streams from altered behavior of small groups of birds; broad traffic patterns can change from minor disruptions; epithelial cells can adopt different modes of migration with varying ratios of cell identities or levels of force-generating proteins within cells (Guttal and Couzin, 2010). Cell collectives, namely, grouped epithelial cells, play crucial roles in development,

functions, and maintenance of a variety of organs and tissue structures (Blanpain *et al.*, 2007; Park *et al.*, 2016). A key aspect of epithelial cells' function is collective migration wherein cells at the leading edge communicate with followers to enable navigation of epithelial collectives through heterogeneous environments (Khalil and Friedl, 2010). Migrating cell collectives have been physically described in terms of active matter that can transition across solid, fluid, or gas phases according to biochemical cues or the disease state of cells

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Summary statement: Topological obstructions in extracellular matrix soften epithelial cells and slow their migration. Cellular obstruction sensing requires an optimal balance of protrusions and cell–cell adhesions.

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Abbreviations used: AFM, atomic force microscopy; F-actin, filamentous actin; GTP, guanosine triphosphate; MCF10A, Michigan Cancer Foundation-10A; MCSs, Monte Carlo Steps; MDCK, Madin-Darby Canine Kidney; MSD, Mean squared displacement; PDMS, Polydimethylsiloxane; WT, wild-type.

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(Angelini *et al.*, 2011; Tambe *et al.*, 2011; Kim *et al.*, 2013). Numerous studies have also shown that extracellular matrix (ECM) properties influence collective cell migration from ECM stiffness and chemokine gradients to physical confinement and surface topography (Ng *et al.*, 2012; Kim *et al.*, 2014; Nasrollahi and Pathak, 2016). Tissue and organ surfaces within the body (e.g., bone, lung, and skin) are not always smooth and have topological imperfections that can obstruct planar migration (Saw *et al.*, 2017; Al-Izzi and Morris, 2021). It remains unclear how collectively migrating cells negotiate topological obstructions on matrix surfaces.

Collective cell migration is a multiscale process, regulated by mechanosensitive proteins, cell-ECM adhesions, cell-cell adhesions, and large-scale force propagation (Friedl *et al.*, 2012). In general, cells at the leading edge of migration, the leader cells, migrate outward such that biochemical and mechanical signals are efficiently propagated throughout the epithelial sheets via cell-cell adhesions (Treat *et al.*, 2009; Vishwakarma *et al.*, 2018; Hino *et al.*, 2020). This force transmission from leader to follower cells operates with such sensitivity that the entire monolayer can respond to mesoscale ECM stiffness gradients that change over just tens of microns (Sunyer *et al.*, 2016). In this macroscale coordination, dynamic cytoskeletal remodeling regulates leader cell protrusions and shape of cells within the monolayer (Te Boekhorst *et al.*, 2016), while adherens junctions enable cell-cell communication across epithelia (De Pascalis and Etienne-Manneville, 2017). Recent work has discovered that cell shape is an important indicator of the fluid-like behavior of the monolayer, defined by mesoscale phenomena known as “jamming and unjamming” (Park *et al.*, 2015). In general, unjamming refers to a fluid-like and promigratory cellular state, which is characterized by elongated morphology and increased migration speed. By contrast, cell jamming typically defines a solid-like cellular state, with rounded morphology and limited forward migration (Park *et al.*, 2015). The jamming–unjamming or solid–fluid transitions at cellular and population scales have been modeled as transitions between energy states, which have been implemented in agent- and lattice-based models (Alert and Trepat, 2020). In such models, generally, energy contributions corresponding to cell–cell adhesions and active motile forces are assumed to be antagonistic, which ensures that strengthening cell–cell adhesions reduce cellular migratory forces and vice versa (Schwarz and Safran, 2013; Bi *et al.*, 2015; Alert and Trepat, 2020). Thus, the energy barrier between jammed and unjammed states can be viewed as a difference between these two cellular energy levels. Taking into account random propulsion forces within the monolayer and cortical tension, cellular state transitions can be determined through the shape index of the cells, a ratio of the cell perimeter and area (Bi *et al.*, 2016). This physical modeling technique allows for visualization of how microscale changes in parts of a cell or whole cells can yield varying outcomes on mesoscale and macroscale collective migration behaviors.

In diseases such as asthma and cancer (Acerbi *et al.*, 2015; Park *et al.*, 2015; Ilina *et al.*, 2020; Mitchel *et al.*, 2020), biological state of cells and mechanics of the ECM evolve and both of which can alter collective cell migration. Previous studies have shown that properties such as ligand density, geometry, and substrate stiffness all have pronounced effects on single-cell migration (Pathak and Kumar, 2011). In general, cells tend to generate higher cellular forces and move faster on surfaces of optimal stiffness, confinement, and ligand density (Pelham and Wang, 1997; Peyton and Putnam, 2005; Pathak and Kumar, 2012; Xi *et al.*, 2017). Because collective cell migration requires integration of cell-level force generation and multicellular signal propagation, which operate at multiple lengths and timescales (Treat and Sahai, 2018), it is likely

that microscale changes in the ECM can result in large-scale epithelial responses. For example, longer collagen fibers can result in sustained collective cell streaming due to increased cell–cell correlation (Sarker *et al.*, 2019). When physical defects (<50 μm in size) are introduced in the basement membrane mimicking surfaces, healthy epithelial collectives can undergo disease-like epithelial–mesenchymal transition and lateral invasion due to long distance propagation of degraded collagen-IV (Walter *et al.*, 2018). Collective epithelial populations can also undergo epithelial–mesenchymal transition when migration on substrates of micropillars (Wong *et al.*, 2014). Through development and disease, the ECM can degrade and remodel, which results in varied and heterogeneous regional topology within the matrix (Lu *et al.*, 2011). Thus, it is important to further study how cell collectives sense and respond to topological heterogeneities, as well as how these signals are communicated across cell population. Because cell collectives have been shown to move similar to solids, fluids, or gases, depending on their cell type and environment (Angelini *et al.*, 2011; Kim *et al.*, 2013), it remains unclear how migrating cell collectives respond to topological obstructions in the matrix. Although it is expected that densely spaced matrix obstructions would disrupt cell–cell communication and force propagation, it is somewhat unpredictable whether such obstruction sensing varies with levels of cell–cell adhesions or migratory forces.

Collective cell migration phenotypes depends on cellular forces, multicellular cooperation, and environment properties, which span a wide range of length scales. In this study, we broadly define microscale processes as those operating on the subcellular or single-cell scale (<10 μm). We measure macroscale responses in leading edge structure and net monolayer migration (>100 μm). Finally, we define mesoscale properties in the intermediate length scale that covers small groups of cells or matrix structures in the 10–100- μm range. To understand how mesoscale obstructions in matrix topology affect collective cell migration, we microfabricate polydimethylsiloxane (PDMS) substrates with nonflat and stump-like geometries. Both the number and the orientation of the stumps were adjusted to study the effects of density and orientation of obstructions on cell migration. We then performed collective cell migration analyses, single-cell-level atomic force microscopy (AFM), and cell shape analyses to fully understand the sensing, transmission, and response of cells to matrix obstructions. We also developed a computational model to understand how the mesoscale environmental changes result in microscale cellular modifications and macroscale obstructed collective migration. We discovered that dense matrix obstructions in the path of migrating healthy epithelia cause a slowdown in the forward migration, loss of directionality, reduced cellular fluidity, and cell-level softening, all of which result in disordered leading edges. This sensitivity to obstructions is differentially regulated by cell protrusions and cell–cell adhesions. Overall, our results reveal that mesoscale matrix obstructions can cause multiscale physical changes in collective migration of cell populations according to their ability to generate migratory forces and perform cell–cell communication.

RESULTS

Engineered matrix obstructions in the path of epithelial monolayers

To understand whether mesoscale obstructions in matrix topology can alter how epithelial cells migrate collectively, we use soft lithography to microfabricate PDMS substrates with defined stumps. We began by using rectangular stumps of varying aspect ratio and edge length between 25 and 500 μm (Supplemental Figure S1). From these trials, we observed more migration hindrance when the

perpendicular dimension of the obstruction is greater than its parallel dimension. However, such wide variations in shapes, sizes, and density of obstructions made it difficult to interpret their influences on migration phenotypes. Therefore, to isolate effects of obstruction orientation and density, we decided to keep dimensions of the stumps constant at $100 \times 25 \mu\text{m}$ (Figure 1A) and placed them parallel or perpendicular relative to the direction of cell migration. As a matter of terminology, in this study, we will refer to these topological features as “obstructions” when discussed from the standpoint of cells, and as “stumps” in the context of matrix properties. We varied spacing between individual stumps such that “sparse” and “dense” obstructions are 50- and $150\text{-}\mu\text{m}$ apart, respectively. We also varied the orientation of stumps relative to the direction of cell migration, such that they present either short obstructions of $25 \mu\text{m}$ in the case of parallel orientation or long obstructions of $100 \mu\text{m}$ in the case of perpendicular orientation (Figure 1B). These dimensions were chosen such that the whole cell body could interact with stumps. From time-lapse cell migration imaging, we computed correlation length over which one cell's motion is correlated with another in the collectively migrating monolayer. Accordingly (Figure 1C), dense spacing of $50 \mu\text{m}$ is roughly around the average correlation length in most cases, which is intended to disrupt cell-cell cooperation and correlated collective migration. Sparse spacing of $150 \mu\text{m}$ is larger than the higher end of correlation length, which is intended to maintain cell-cell cooperation. The substrate was designed such that epithelial cell monolayer is first seeded in an open space, without stumps, and allowed to migrate outward before running into obstructions presented by the stumps (Supplemental Figure S2; detailed in Methods). After coating the PDMS substrates with fibronectin, we cultured MCF10A human mammary epithelial cells. Through live imaging, we studied cell interaction with stumps and various collective migration phenotypes over time through these varying obstruction courses. For initial understanding, these measurements are first compared against a control condition of PDMS substrates with stump-free “flat” space; however, the key focus of the study is to compare the effects of sparse versus dense and parallel versus perpendicular obstructions on collective cell migration.

Multiscale disorder in obstructed collective cell migration

To measure the response of collectively migrating cells to obstructions presented by matrix topology, we performed time-lapse imaging for more than 24 h and analyzed trajectories of individual MCF10a cells within the epithelial monolayer. We observed that epithelial monolayers moving through denser obstructions, particularly those in a parallel orientation, had more ruffled leading edges and overall shorter distance traveled than sparse-stumps or stump-free (flat) substrates. As shown in Figure 1D, the epithelial monolayer migrating through open space or sparse obstructions progressed over time without significant backward retraction. However, when migrating through dense obstructions, the leading edges (tracked at 6-h intervals) intersected with each other, indicating their dynamic frontward and backward movement as they navigated around each obstruction. As a result, the leading edge became chaotic and ruffled when moving through many obstructions, particularly in the dense condition (Figure 1D, each time frame visualization provided in Supplemental Figure S3). To test whether cell proliferation could play a role in obstruction sensing, we arrested cell cycle progression in MCF10A cells by treating with itraconazole (Wang et al., 2017) and showed that the migration differences between dense and sparse obstructions still persisted despite proliferation inhibition (Supplemental Figure S4).

Macroscale disruption in the leading edge integrity caused by dense obstructions likely alters cell-scale migration behaviors. To investigate this, we tracked and plotted the speed of all cells that form the collective monolayer. As visualized in Figure 1E, on stump-free, open substrates, cells were homogeneously fast, regardless of their distance from the leading edge into the colony. However, the presence of obstructions (sparse and dense) led to more heterogeneous and overall slower migration speeds of cells within the monolayer, with faster cells at the leading front. Plotting the average migration speed across samples (Figure 1G) confirmed these observations of more heterogeneous (see bigger error bars in Figure 1G), yet slower average migration speeds in the case of dense obstructions, compared with sparse or no obstructions. Throughout the full 30-h duration of tracking, cell migration speed in all conditions remained steady (Figure 1H). Among these tested substrates, cell migration was the fastest on substrates with sparse-perpendicular stumps or those without any stumps. In other words, cell migration was the fastest when the contact area of obstructions in the direction of migration was the smallest. When sparse stumps were oriented parallel to the direction of migration, the migration speed slowed. Dense stumps led to the slowest migration overall, with similar speeds for both orientations (Figure 1G).

Because obstructions reduce individual cell speeds in our data, we wondered whether the physical impact of obstructions on cells reduces fluidity of the collective. To understand this, we measured mean square displacement (MSD) relative to varying time intervals (Δt) averaged for all cells (Figure 1I) and overall time-averaged diffusivity obtained from MSD (Figure 1J), as done previously (Park et al., 2015). In the linear MSD plot (Figure 1I), lower slope and unsteady relationship between MSD and Δt for denser obstructions indicate more disordered migration. However, diffusivity of cellular migration did not appreciably change across conditions (Figure 1J). To better understand how the loss of collective cell fluidity arises, we measured the directionality of individual cells, defined as a product of direction autocorrelation and cell displacement (see *Materials and Methods*). According to this definition, higher directionality indicates more persistent and ordered migration toward the leading edge. We found that cell migration directionality reduced with increasing density of obstructions (Figure 1K). In the case of sparse obstructions, there was a gradient of higher directionality toward the leading edge cells (Figure 1F). However, in dense obstructions, we noted that the spatial distribution of directionality was somewhat homogeneous across the monolayer. Together, these results show that mesoscale matrix obstructions reduce microscale directionality and migration speed of cells within the epithelial monolayer, which results in macroscale loss of fluidity and chaotic leading edge shapes.

Cellular softening around dense obstructions

According to our results, matrix obstructions slow collective cell migration through mesoscale disorder. Do microscale mechanical properties of cells change as they navigate these topological obstructions? Although cell-level tractions in epithelial populations have been connected to varying jamming and unjamming phenotypes of collective cell migration (Tambe et al., 2011; Park et al., 2015; Mitchel et al., 2020), cell stiffness may not be directly correlated to tractions. Indeed, while cancer cells generate higher tractions, they have been found to be softer than healthy cells (Xu et al., 2012; Network et al., 2013). Because cells are known to soften while moving through confining environments (Rianna et al., 2020), changes in cell stiffness could occur because cell body must physically interact with the stumps to move through matrix obstructions. To our knowledge, cell stiffness in collectively migrating monolayers

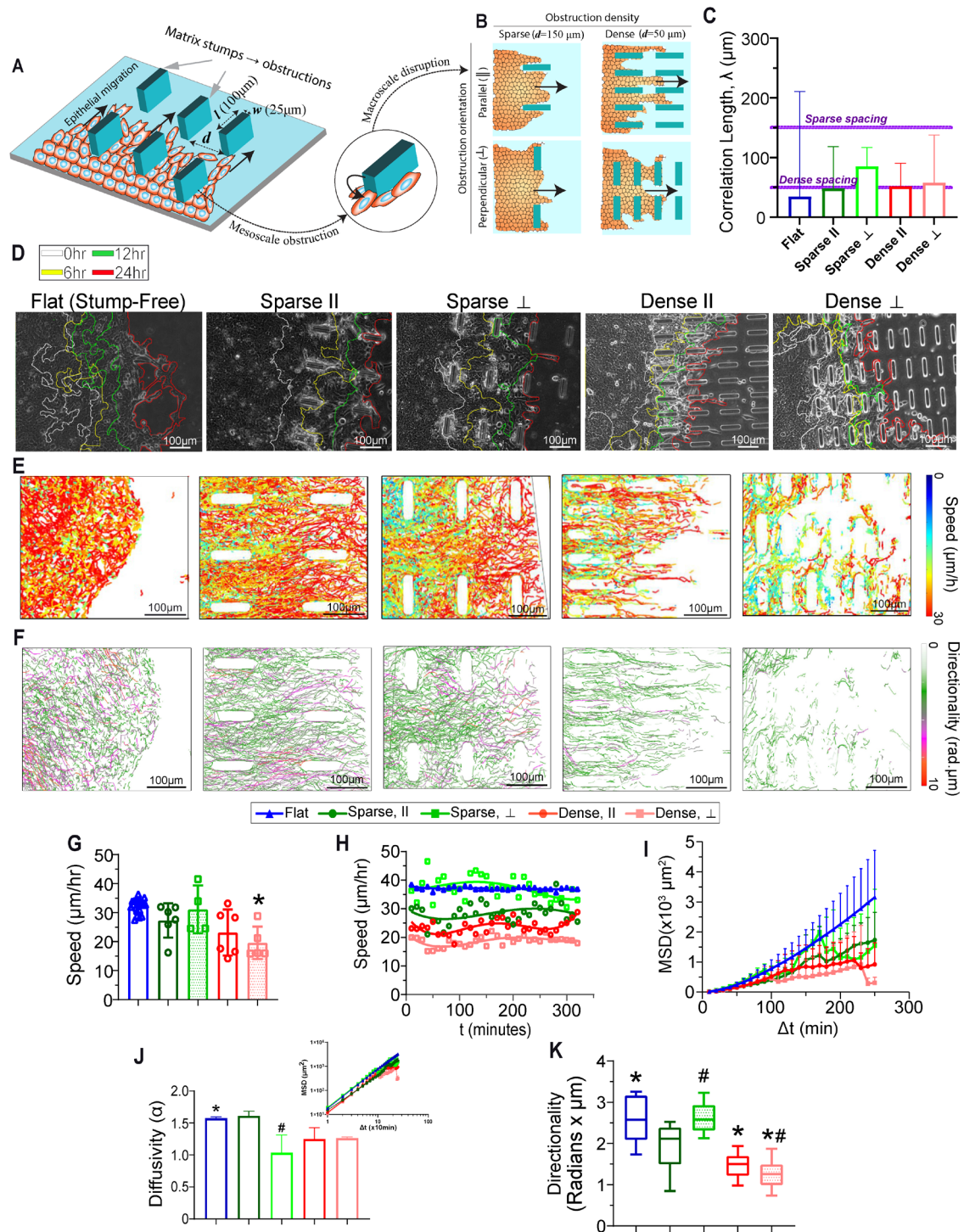


FIGURE 1: Dense obstructions reduce collective cell migration speed and directionality. (A) Schematic describing collective epithelial migration on engineered substrate with rectangular stumps of dimensions $l = 100 \mu\text{m}$ and $w = 25 \mu\text{m}$, which serve as mesoscale matrix obstructions. (B) Varying spacing and orientation of stumps generate four different conditions for obstructions in the path of collective cell migration. (C) Average correlation length (λ) of epithelial migration on substrates with stumps of varying spacing and orientation. (D) Representative phase contrast images of migrating MCF10A epithelial cells in different substrates at final time point (24 h after cell seeding), with leading edges annotated at previous time points in 6-h intervals. In images, cells migrate left to right. Migration trajectories of individual cells within epithelial monolayers are plotted with color coding for time-averaged (E) speed and (F) directionality. (G) Migration speed averaged over all cells per condition over 3 d of migration. Error bars = SD. (H) Temporal variation in average migration speed for each substrate condition; points represent average across all cells for a specific time point; and lines are polynomial fits across time points. (I) Plot of mean squared displacement (MSD) versus duration of time interval Δt . (J) Cell-level diffusivity (α) calculated from the accompanying MSD- Δt log-log plot. (K) Directionality of cell migration averaged over all cells per condition over 3 d of migration. Error bars = SEM. Significant difference annotated as * $p < 0.05$ for comparison against the sparse condition, and # $p < 0.05$ for comparison against the parallel condition. $N \geq 4$ fields of view in all plots.

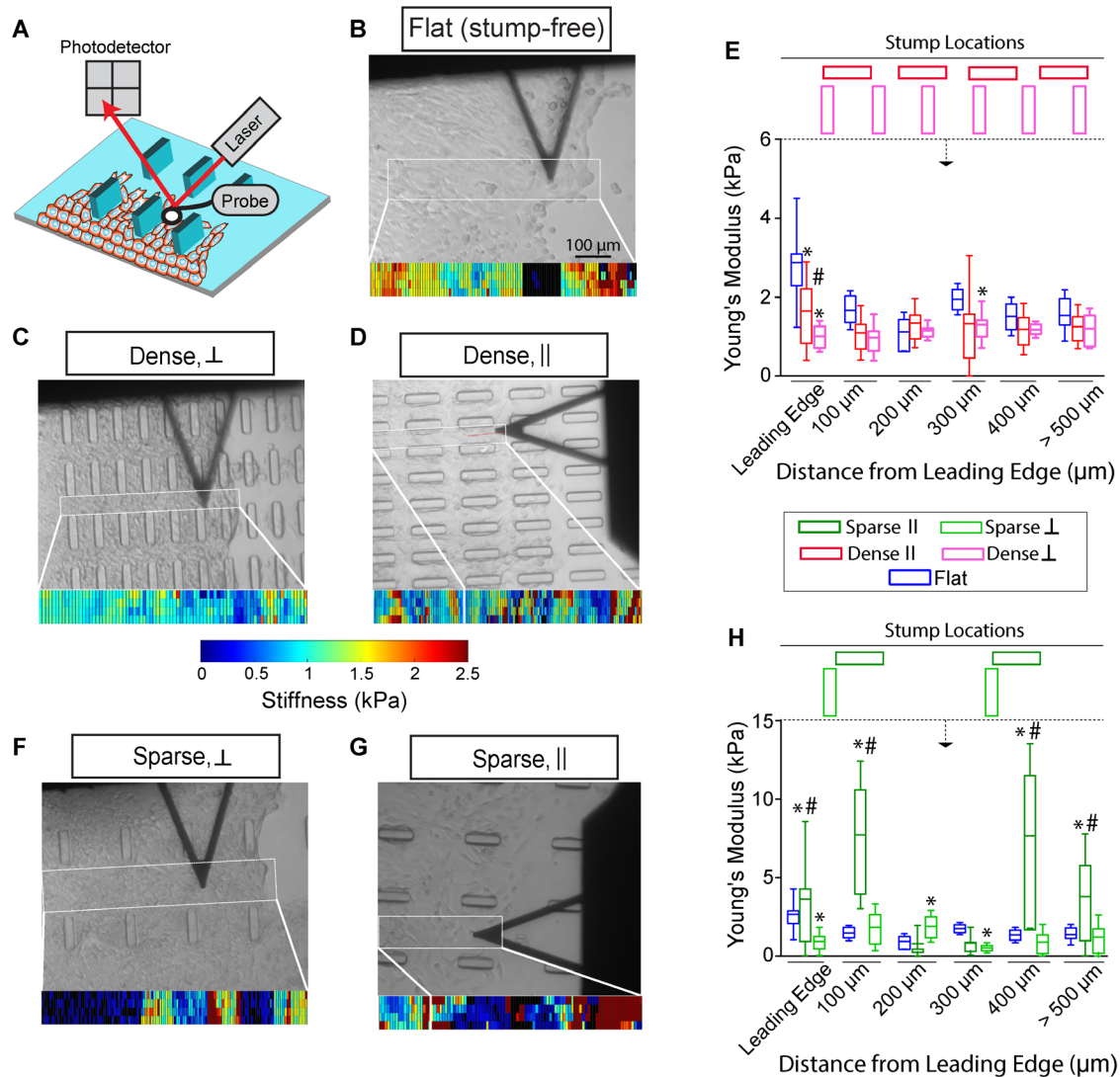


FIGURE 2: Cell stiffness responds to mesoscale obstructions. (A) Representative diagram of how cell stiffness was determined using AFM (see *Materials and Methods*). Representative heat maps of stiffness scans along with corresponding brightfield images for (B) flat, (C) dense-perpendicular, and (D) dense-parallel conditions. White boxes and bars indicate the corresponding region where the heat map aligns with the cell image. Scale bar = 100 μm. (E) Cell stiffness by distance from the leading edge for flat and dense obstructions with both parallel and perpendicular orientations. Bars at the top of the graph indicate approximate location of stumps for their given orientation. Box represents 25–75th percentile, with a mean line and SD whiskers. Representative heat maps of full stiffness measurement runs with corresponding brightfield images for (F) sparse-perpendicular and (G) sparse-parallel stumps. White boxes and bars indicate the corresponding region where the heat map aligns with the cell image. Scale bar = 100 μm. (H) Cell stiffness by distance from the leading edge for flat and sparse obstructions with both parallel and perpendicular orientations. Bars at the top of the graph indicate approximate location of stumps for their given orientation. Box represents 25–75th percentile, with a mean line and SD whiskers. Significant difference annotated as $*p < 0.05$ for comparisons against flat substrates, and $\#p < 0.05$ for comparison between parallel and perpendicular conditions. $328 \geq N \geq 21$ number of AFM scans per position from three samples for each substrate condition.

has not been measured. We assessed changes in cell stiffness through AFM. Briefly, substrates were loaded onto the AFM stage so that the stumps ran parallel to the AFM cantilever to ensure that the AFM probe interacted with the cells without colliding with the stumps (Figure 2A). We used AFM probes with spherical tips of 4.5-μm diameter and indented cells over 0.25-μm depth to acquire force displacement curves, which were processed using a modified Hertz model to calculate microscale cell stiffness (details in Methods, and Supplemental Figure S5B). Testing began at the leading edge of the monolayer, and full runs were taken in 100-μm increments

until the region of the monolayer before the first stump was reached (Supplemental Figure S5B). Each section was selected such that measurements were taken as close to the stumps without the cantilever touching the stumps and thereby skewing the stiffness measurements.

As a control, we first measured the stiffness of cells that form the epithelial monolayer on flat substrates by scanning from the leading edge into the monolayer. As shown in the representative monolayer and the spatial AFM scan, cells at the leading edge were stiffer than the cells within the monolayer, with average Young's moduli reducing

from ~2.8 to 0.5 kPa (Figure 2B), that is, leader cells were much stiffer than follower cells embedded within the epithelial monolayer. Although this variation in leader–follower cell stiffness has not been shown before, it is not entirely surprising, given the known rise in mechanosensitive proteins, phosphorylated myosin light chain (Ng *et al.*, 2012), and yes-associated protein (Sarker *et al.*, 2019) toward the leading edge. To understand the effect of matrix obstructions on cell stiffness, we performed AFM on cell monolayers that had already migrated into dense stumps. According to the representative spatial maps for cells around dense stumps, cell stiffness was generally lower compared with the flat substrates (Figure 2, C and D). Going from the leading edge into the monolayer, cell stiffness was relatively homogenous and consistently lower in the case of dense-perpendicular obstructions (Figure 2C), compared with more distinct patches of soft and stiff cells in parallel obstructions (Figure 2D). For these three substrate configurations, we repeated AFM measurements for at least 3 different samples and averaged the cell stiffness, which was binned relative to distance from the leading edge, as plotted in Figure 2E (location of the stumps is denoted in the background). On average, cell stiffness at the leading edge was the highest in the epithelial monolayers on control (more intact) flat substrates (~2.5 kPa) and reduced by almost half around dense parallel obstructions (~1.4 kPa), with even further reduction in case of dense perpendicular obstructions (~0.8 kPa). These results (Figure 2, B–E) show an overall cell softening, particularly at the leading edge, when epithelial cells are forced to navigate dense matrix obstructions.

Next, we sought to test changes in the stiffness of cells moving collectively through sparse stumps, an intermediate presentation of obstructions. In sparse–perpendicular obstructions, as shown in the representative AFM map in Figure 2F, we observed patches of high cell stiffness (~2.5 kPa), but overall cell stiffness across the epithelial cell monolayer seemed lower than that on flat substrates. When we repeated these measurements in sparse–parallel obstructions, heterogeneity in cell stiffness across the monolayer was further accentuated, with “stiff patches” of much higher cell stiffness compared with any other condition (Figure 2G). Given that dense obstructions softened cells, higher cell stiffness in the case of intermediate stump density, although only in the case of parallel orientation, was a surprising result. To confirm these trends and find a potential explanation for these results, we repeated these measurements 3 more times on different monolayers. The overly stiff patches were localized in epithelial regions between parallel stumps. Upon plotting cell stiffness as a function of distance from the leading edge and relative to the location of stumps, the average cell stiffness sharply increased, to approximately three-fold, in regions between the parallel stumps. In all other regions, cell stiffness values were similar to those in flat or sparse–perpendicular substrates (Figure 2H).

According to these results, although dense obstructions broadly softened cells within epithelial monolayer (Figure 2, C–E), sparse–parallel stumps led to a complex effect on cell stiffness (Figure 2, G and H). Overall, there are two key takeaways from these AFM measurements: (1) on flat substrates, leader cells are significantly stiffer than follower cells; and (2) dense obstructions soften the leader cells and thus make them behave like follower cells in terms of their stiffness.

Loss of cell shape fluidity due to matrix obstructions

As epithelial colonies move through and around matrix obstructions, individual cells within the colony lose directionality and speed in their migration (Figure 1), along with general softening and meso-scale spatial heterogeneity in cell stiffness relative to distance from

the leading edge and density of obstructions (Figure 2). Following these trends down the length scale—from the leading edge roughness at the scale of hundreds of microns to migration and stiffness at the scale of tens of microns—we next sought to measure cytoskeletal and cell-level changes due to obstructions. Given the known importance of actin polymerization in protrusions that propel migration (Gardel *et al.*, 2008; Petrie *et al.*, 2009) and the actin-rich cytoskeleton that provides mechanical stability to the cell (Kumar *et al.*, 2006), we stained with phalloidin to measure F-actin intensity. We found higher actin expression near the leading edge on all substrates, albeit with some obstruction-dependent distinctions (Figure 3A). In denser obstructions, we noted that areas of high actin intensity were smaller and localized closer to the leading edge. By contrast, cell monolayers migrating through sparse stumps showed broader regions of high F-actin intensity near the leading edge. After averaging across multiple samples, we found a significant reduction in actin expression in dense obstructions compared with that in sparse obstructions (Figure 3B). We noted that the highest actin expression was in sparse–parallel obstructions. Thus, the unexpectedly high cell stiffness in this condition (discussed previously in Figure 2) could potentially be due to the increased actin expression (Figure 3B), which remains unexplained.

In previous work, various collective cell migration phenotypes have been described in terms of jamming and unjamming within epithelial colonies derived from healthy or diseased states, that is, in asthma and breast cancer (Park *et al.*, 2015; Iliina *et al.*, 2020). In these cases, the “unjammed” cells move in a fluid-like state, characterized by high shape index, defined as $P/\sqrt{A}$ (P is the perimeter and A is the area of cells that form the colony). In our system, the same healthy epithelial colony alters its migration phenotype, mechanical properties, and cytoskeletal composition due to the presence of matrix obstructions—fast and fluid-like migration on substrates with sparse obstructions changes to more frustrated migration through dense obstructions (Figure 1). Notably, these multiscale changes in migration occur without any prescribed biochemical, genetic, or pathological alterations in cells. The loss of migration fluidity in dense obstructions was accompanied by physical softening and loss of cytoskeletal actin (Figure 2, Figure 3, A and B). To understand whether cell shape fluidity follows these trends, we measured the shape index of the cells in epithelial monolayers migrating around the stumps. As plotted in Figure 3C and visualized in Figure 3D, the shape index of cells around sparse stumps was higher compared with dense stumps, which indicates that cell shape fluidity reduces due to cell interaction with densely packed obstructions. Here, the cell shape index was the highest in the case of sparse–parallel obstructions, which is consistent with the trends we observed previously for cell stiffness and actin intensity.

Lattice-based modeling to understand obstructed collective cell migration

Physical understanding of collective cell migration has been developed through direct cellular measurements in concert with computational models based on cell-generated forces, cell–cell cooperation, and cell–ECM adhesions (Alert and Treppe, 2020). By combining physics-based and phenomenological observations, cells can be treated as agents that move according to a net energy minimization criterion in which the rules of cohesion and scattering are incorporated in terms of intercellular potentials (Camley and Rappel, 2017; Copenhagen *et al.*, 2018). In dynamic vertex-based models, cell migration is generated from nodal movements that minimize local energy, and fluidic collective motion is captured through dynamic cell-shape changes (Mitchel *et al.*, 2020). Similarly, lattice-based models

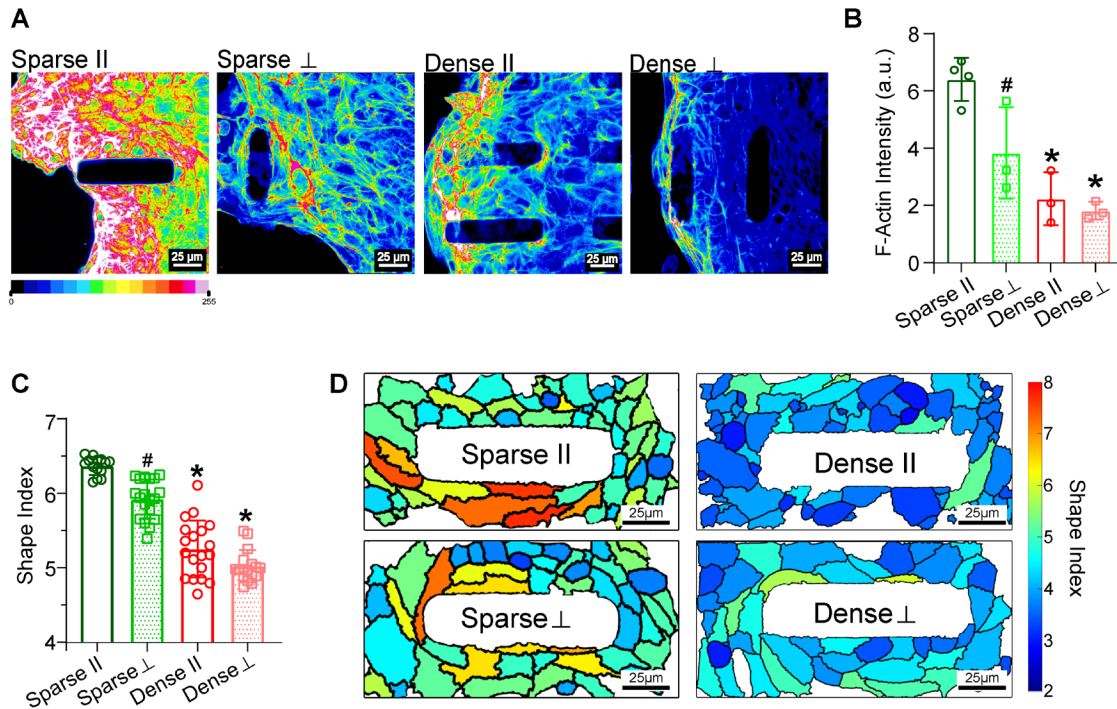


FIGURE 3: Dense obstructions reduce cell shape fluidity. (A) Representative images showing F-actin intensity (phalloidin) in cell monolayer around dense or sparse obstructions with parallel or perpendicular orientations. (B) Average F-actin intensity normalized by area occupied by cells across. Error bars = SD, $N \geq 3$ samples. (C) Average cell shape index and (D) representative cell shapes color-coded with cell-level shape index values. Error bars = SD, $N \geq 10$ regions of interest around stumps. Significant difference annotated as $*p < 0.05$ for comparison against the sparse condition, and $\#P < 0.05$ for comparison against the parallel condition.

generate single and collective cell migration such that any movement minimizes the net system energy (Graner and Glazier, 1992; Fortuna *et al.*, 2020). Such approaches have led to new insights regarding how any given biological system differs from the original biophysical system (Mogilner and Odde, 2011; Camley and Rappel, 2017). Along these lines, to understand how obstructions alter collective cell migration, we first develop a simple lattice-based model for collective cell migration and apply as few rules as possible to capture experiments. To this end, we note a key experimental observation that had not been shown before this study: dense obstructions induce follower-like cell response in terms of reduced cell stiffness, actin expression, and migration phenotypes in the form of disordered leading edge and reduced collective migration speed and directionality.

We developed a Cellular Potts model (Graner and Glazier, 1992) for collective cell migration in the presence of obstructions (as detailed in *Materials and Methods*). Briefly, active energy for cell migration depends on two factors: (1) cell type-dependent contractile energy ϕ that resists protrusions for all cells in a population, and (2) additional protrusive energy of leader (α_L) and follower (α_F) cells that changes according to cells' location within the migrating population. Here, higher contractile energy ϕ reduces net protrusions for the whole cell population, which can be used to define different cell types. On flat surfaces (without obstructions), leader cells pull on follower cells to generate collective migration (Figure 4A). According to our experiments, it is likely that protrusions in leader cells are suppressed by obstructions. To capture this, when leader cells come in contact with obstructions, our model assumes that active forces generated by a leader cell are reduced through a rise in contractile energy ϕ and reduction in protrusive energy (α_L), as described in Figure 4B. Given the known macroscale response of obstructions on

collective migration, we also assume that signals from leader cells are immediately propagated throughout all cells in the monolayer.

We simulated a monolayer of about 4000 cells migrating into sparse or dense obstructions in parallel or perpendicular orientations. The front-rear polarity in each cell is randomly chosen between -90° and 90° angles, with reference 0° angle in rightward direction. Thus, the entire monolayer moves forward into the region with obstructions. We performed simulations of cell migration through obstructions over 4000 Monte Carlo Steps (MCSs) equivalent to 3 d. All monolayers started at the same point, outside the region with obstructions, and the monolayer at the final time point is visualized in Figure 4C for sparse and dense obstructions with parallel orientations (results for perpendicular orientations are shown in Supplemental Figure S6). Here, dense obstructions reduced net monolayer translocation and average migration speed, which is broadly consistent with experiments. As visualized through color-coded cells (Figure 4C), the net energy cost (H) of cell migration increases with higher density of obstructions. Also similar to experiments, the leading edge integrity was more compromised in dense obstructions, because of higher heterogeneity in net energy at the leading edge. These simulations indicate that our simple modeling assumption of suppressed leader cell function upon contact with obstructions is enough to capture key experimental observations.

Next, we sought to understand how cells with inherently different abilities to generate migratory protrusions and perform communication via cell-cell adhesions would sense obstructions. To this end, we repeated our simulations for varying cell-cell adhesions (J) and protrusions (regulated by parameter ϕ , as discussed previously). We first simulated weak ($J = 0.2$) and strong adhesions ($J = 5$) while keeping control protrusions with $\phi = 3$. As shown via color-coded

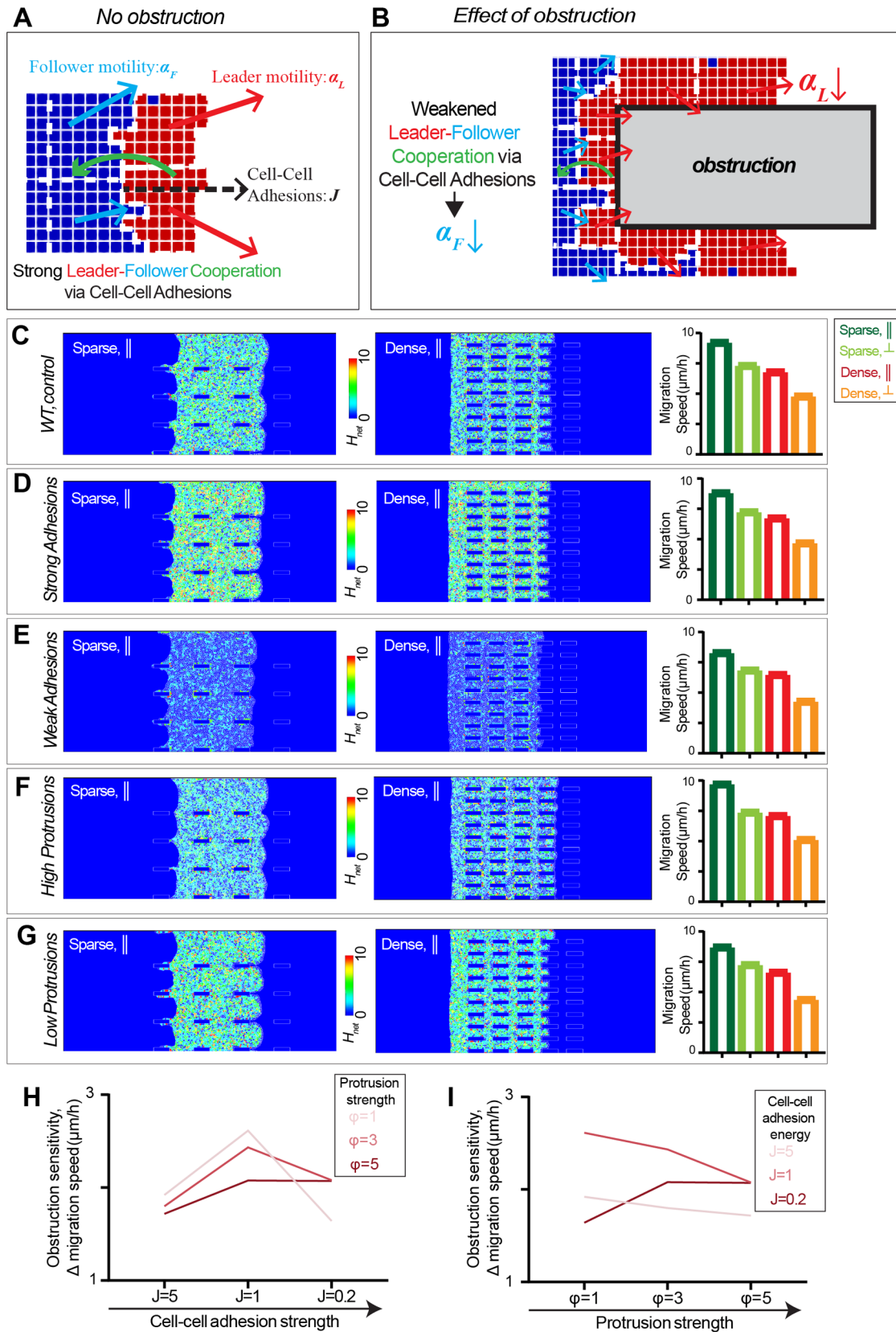


FIGURE 4: Lattice-based modeling of collective cell migration through matrix obstructions. (A) Cellular Potts visualization of a migrating cell colony describing high leader cell motility and leader-follower cooperation. (B) In the presence of obstruction, leader cell motility and leader-follower cooperation are weakened. Simulations are performed and monolayer state is plotted at $t = 72$ h along with bar plot of average migration speed of monolayer leading edges for parallel obstructions for (C) control cell type with moderate adhesions and protrusions, (D) strong adhesions with

cells in Figure 4, C–E (left and middle panels), the net energy cost for migration (H) increased for stronger cell–cell adhesions ($J = 0.2$). We also plotted average migration speed leading edges for all three conditions and found faster cell migration in sparse obstructions (right panels in Figure 4, C–E). In the next set of simulations, we kept control levels of cell–cell adhesions at $J = 1$ and varied parameter ϕ to simulate cell types with high ($\phi = 1$) and low ($\phi = 5$) inherent protrusive abilities. Analogous to the case of stronger cell–cell adhesions (Figure 4D), cells with weak protrusions (Figure 4G) require higher energy cost to migrate, as visualized by cells color-coded for net energy (H). Cells migrated faster in sparse obstructions in these variations of protrusions (right panels in Figure 4, C, F, and G). Upon closer examination, we observed that the levels of difference in migration speeds between dense and sparse obstructions changed with J and ϕ (Figure 4, C–G).

To quantify how different cells respond to obstruction densities, we define “obstruction sensitivity” as the difference in migration speed between sparse and dense obstructions for any given set of cell-specific protrusions or adhesions (ϕ, J). As plotted in Figure 4, H and I, obstruction sensitivity of cells with strong protrusions ($\phi = 1$) is the highest when cell–cell adhesions are of moderate strength ($J = 1$), while both strong or weak adhesions reduce cells’ ability to migrate differently in dense and sparse obstructions. This biphasic dependence of cells’ obstruction sensitivity on cell–cell adhesion strength slightly weakens as protrusions are reduced ($\phi = 3$). For weaker protrusions ($\phi = 5$), obstruction sensitivity loses biphasicity and increases with cell–cell adhesion strength, with no disadvantage for strong cell–cell adhesions. When we plotted the same results against protrusion strength, we observed that obstruction sensitivity reduces with increasing protrusion strength for both weak and strong cell–cell adhesions. Only when cell–cell adhesions are moderate ($J = 1$), obstruction sensitivity increases with protrusion strength, because cells can both migrate fast and perform leader–follower communication to sense obstructions.

These results indicate that cell types with weak or strong cell–cell adhesions are less obstruction sensitive for any level of protrusions, and cell types with moderate cell–cell adhesions exploit higher protrusions to better distinguish between sparse and dense obstructions. Motivated by these modeling predictions, we experimentally test whether variations in cells’ protrusions or adhesions (different cell type or direct perturbations in MCF10A) change their ability to distinguish between dense and sparse obstructions.

Reduced obstructions sensing by cells with higher intercellular E-cadherin

Our simulations predicted that cells with stronger cell–cell adhesions and lower protrusions would be less sensitive to obstructions. To test this, we repeated migration measurements in Madin–Darby canine kidney (MDCK) cells, another epithelial cell line that forms more compact epithelial monolayers than MCF10A. We imaged for E-cadherin in both these cell lines (Figure 5, A and C) and found that localization of E-cadherin at cell–cell junctions was higher in MDCK cells than in MCF10A cells in all five substrate conditions (Figure 5, B and D). For MCF10A cells, perpendicular orientation of obstruction

was more disruptive for cell–cell adhesions (Figure 5B). When migrating through dense obstructions, MDCK cells showed higher E-cadherin expression at cell–cell junctions (Figure 5D), which could indicate their arrested migration through matrix obstructions. We performed live time-lapse imaging of MDCK cells migrating on substrates with varying configurations of obstructions, as shown previously (Figure 1) for MCF10A cells. We observed that leading edges of MDCK cell monolayers were less disrupted and ruffled (Figure 5E) than those of MCF10A cells (Figure 1) when moving through dense obstructions, indicating that MDCK populations are less affected by obstructions. As shown in Figure 5E, the MDCK monolayers did not significantly advance in the 24-h period of migration on substrates of any obstruction configurations. Consistent with this observation, the migration speed of individual cells within the MDCK monolayers was slower than that of MCF10A cells and did not significantly vary with density or orientation of obstructions (Figure 5F). These results indicate that rise in cell–cell adhesions or cell types with stronger epithelial characteristics could be less sensitive to matrix obstructions than that in moderate cell–cell adhesions.

Rac inhibition slows cell migration and suppresses obstruction effects

According to our simulations, microscale suppression of protrusive leaders reduces obstruction-sensitive collective migration (Figure 4). We have also shown that cells moving through dense obstructions lose their shape fluidity, cytoskeletal structure, and stiffness (Figures 2 and 3), generating macroscale reduction in cell migration speed and compromised leading edge integrity (Figure 1). According to our data, cells sense and respond to matrix obstructions on a cytoskeletal level, particularly F-actin expression and cell stiffness. Furthermore, the loss of directionality in dense obstructions indicates that front–rear polarity within the cells may be disrupted as they move around stumps. Given the known role of the small GTPase Rac in actin polymerization and cell polarity in directed cell migration (Ridley *et al.*, 2003; Pankov *et al.*, 2005), we argued that inhibition of Rac could reduce the ability of epithelial cells to sense the density of obstructions. To test these predictions, we performed pharmacological Rac inhibition by treating cells with NSC23766 and monitoring their migration for 24 h. From time-lapse cell-migrating imaging, we noticed a clear change in migration phenotype through dense obstructions—leading edges consistently advanced over time (Figure 6A, each time frame visualizations provided in Supplemental Figure S7) without forming chaotic geometries or backtracking that we had observed in the untreated wild-type (WT) case (Figure 1). Leading edge advancement through sparse obstructions was still higher than through dense obstructions; however, the difference between the two conditions was much less obvious compared with the WT condition. Here, stump orientation did not seem to elicit a clear difference in cell migration phenotypes.

To quantify and visualize the spatial distribution of migration speed on a cellular scale, we plotted cell migration trajectories and color-coded them with their average speeds (Figure 6B). From these speed colormaps, it became more clear that when migrating through sparse obstructions, cells near the leading edge were faster

moderate protrusions, (E) weak adhesions with moderate protrusions, (F) high protrusions with moderate adhesions, and (G) low protrusions with moderate adhesions. Each cell in the colony is color-coded with net energy H_{net} , whose higher values indicate higher energy cost and thus predict reduced migration. Simulation snapshots for perpendicular obstructions are visualized in Supplemental Figure S6. Plots of (H) obstruction sensitivity versus cell–cell adhesion strength for different protrusion strengths and (I) obstruction sensitivity versus protrusion strength for different values of adhesion strength. Arrows indicate the direction of cellular migration. Error bars = SEM, $N = 3$ simulations.

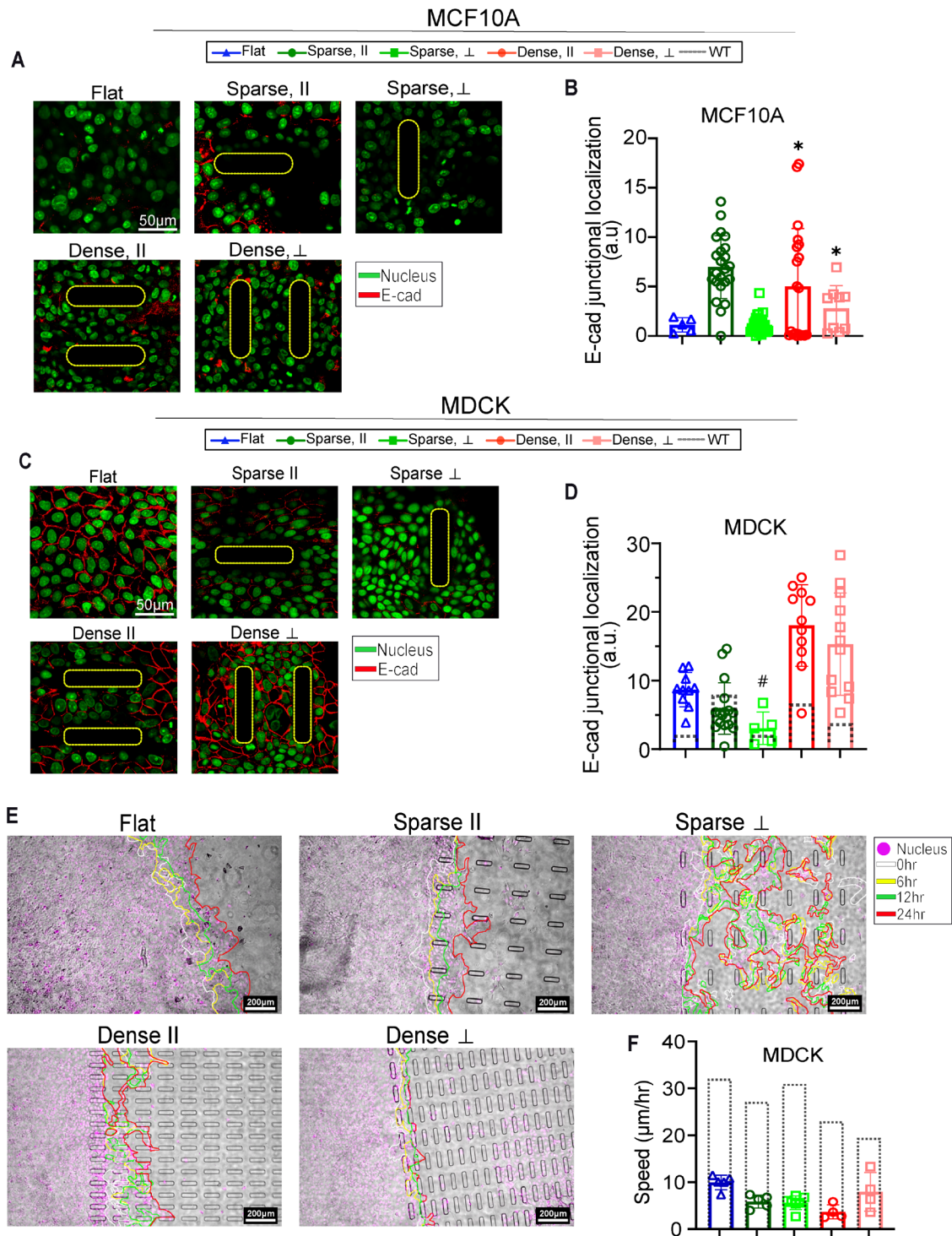


FIGURE 5: MDCK epithelial cells show higher E-cadherin expression and lower obstruction-sensitive migration than MCF10A cells. Representative images of E-cadherin (red) with nucleus (green) in cell monolayer around sparse or dense obstructions with parallel or perpendicular orientation for (A) MCF10A and (C) MDCK. Average area occupied by E-cad near cell-cell interface in each image for (B) MCF10A (D) MDCK. (E) Representative phase contrast images merged with nucleus (magenta) at 24-h time point of MDCK cells in four substrate conditions with varying obstruction spacing and orientation. Leading edges of migrating colonies are annotated for time points at 6-h intervals between 0 and 24 h. (F) Migrating trajectories of individual cells within MDCK epithelial monolayers are plotted with averaged speed. Error bars = SD. Significant difference annotated as * $p < 0.05$ for comparison against the sparse condition, and # $p < 0.05$ for comparison against the parallel condition. $N \geq 5$ fields of view in all plots.

than those in the monolayer interior (Figure 6B; sparse conditions). Thus, long distance communication between leaders and followers may still be occurring on the monolayer scale despite Rac inhibition.

By contrast, although cells migrating through dense obstructions were faster than WT (with more “red” tracks), those fast cells were somewhat randomly distributed throughout the monolayer, without

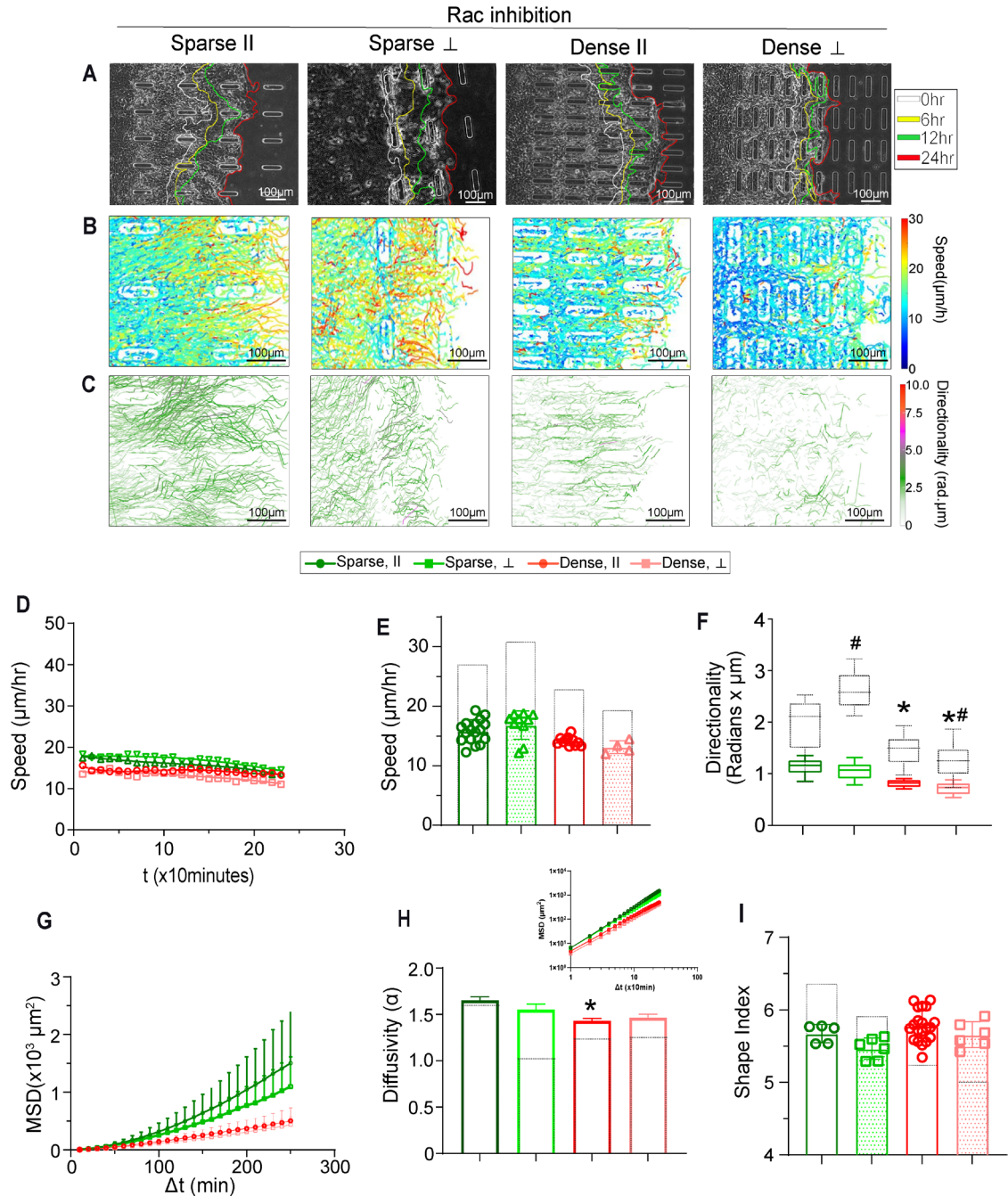


FIGURE 6: Rac-inhibition reduces obstruction-sensitive collective cell migration. (A) Representative phase contrast images at final time point (24 h after cell seeding) of MCF10A epithelial cells treated with Rac inhibitor in four substrate conditions with varying obstruction spacing and orientation. Leading edges of migrating colonies are annotated for time points at 6-h interval between 0 and 24 h. The representative images have the migration direction of left to right. Migration trajectories of individual cells within Rac-inhibited epithelial monolayers are plotted with color coding for time-averaged (B) speed and (C) directionality. (D) Temporal variation in average migration speed for each substrate condition; points represent average across all cells for a specific time point and lines are polynomial fits across time points. (E) Speed and (F) directionality of cell migration averaged over all cells per condition; dotted lines and boxes show data for WT for direct comparisons. Error bars = SD. (G) Plot of mean squared displacement (MSD) versus duration of time interval Δt . (H) Cell-level diffusivity (α) calculated from the accompanying MSD- Δt log-log plot. (I) Average cell shape index plotted for all four substrates. Error bars = SEM. Significant difference annotated as $*p < 0.05$ for comparison against the sparse condition, and $\#p < 0.05$ for comparison against the parallel condition. $N \geq 4$ fields of view in all plots.

any clear localization at the leading edge. These observations indicate that in moving through dense obstructions, epithelial monolayers lose the sense of leaders and followers. Cell migration

speed averaged independently of cell location within the colony stayed consistent over time and was not different between sparse and dense obstructions after Rac inhibition (Figure 6D). When

further averaged across time, although the mean cell migration speed was slightly higher through sparse obstructions than through dense obstructions, it was not a significant difference (Figure 6E). Here, we also noted that Rac-inhibited cells became slower than WT, with a higher decline in speed on substrates with sparse stumps (Figure 6E; Supplemental Figure S8), which is expected, given the known role of Rac in cellular protrusions, polarity, and migration (Pankov *et al.*, 2005; Petrie *et al.*, 2009; Pathak and Kumar, 2013; Sarker *et al.*, 2019). Consistent with this trend, the directionality of migration was also significantly reduced compared with WT, rendering it insensitive to obstruction density (Figure 6, C and F, and stump-free condition included in Supplemental Figure S8). The slight migration advantage through sparse obstructions became more apparent in the relationship between MSD and time interval (Figure 6G). For all substrates with varying obstruction density and orientation, migration diffusivity was around 1.5 (Figure 6H). To analyze cell-shape fluidity, we measured shape index in Rac-inhibited cells around the stumps and found no differences across sparse and dense stumps (Figure 6I). Compared with WT cells, we noted that the shape index decreased around sparse obstructions, but it was rescued in the case of dense obstructions, indicating that the effect of Rac inhibition on cell shape differed for the two cases.

Overall, these results indicate that Rac inhibition reduces the ability of cells to sense and respond to obstructions on the cytoskeletal scale. However, on the macroscale in the whole monolayer, some differences in collective migration phenotypes persisted. In sparse obstructions, there was still a gradient of increasing migration speed from followers to leaders (Figure 6B).

Loss of cell–cell adhesions abrogates macroscale obstruction sensitivity

According to our interpretation of cell stiffness and migration analyses, obstructions make leader cells behave like follower cells (Figures 1 and 2). Moreover, cytoskeletal disruption by Rac inhibition did not completely diminish the ability of cells to distinguish sparse and dense obstructions (Figure 6). In our simulations (Figure 4), the population-wide activation of leader phenotype (without intercellular cooperation) completely diminishes the effects of matrix obstructions on collective cell migration. To further test obstruction sensing, we sought to disrupt cell–cell communication and alter all epithelial cells toward leader phenotype, which would obviate the possibility of leader–follower communication. We targeted α -catenin, which binds to actin and regulates cell–cell adhesions (Robinson and Moberg, 2011). Because α -catenin is a transcription inhibitor, its depletion enhances yes-associated protein activity, induces mesenchymal phenotype, and effectively makes all cells behave like leaders (Daugherty *et al.*, 2014; Vite *et al.*, 2018). We used MCF10A cells with depleted α -catenin (10A- α cat) via shRNAi, developed previously (Loza *et al.*, 2016a), and repeated the migration experiments. Across substrates with sparse and dense obstructions, collective cell migration was chaotic to such a degree that it was difficult to trace leading edges. As visualized in Figure 7A, although the leading edges of the migrating 10A- α cat cell monolayer advanced over time, they were much rougher than WT and there was no real difference across sparse or dense obstructions. According to spatial colormaps of migration speed of individual cells, there were no visible leader–follower gradients in speeds, and fast cells (red tracks) were scattered across the monolayer on all substrates (Figure 7B). This phenotype was particularly unique in the case of dense obstructions, where WT and Rac-inhibited cells were much slower. Indeed, 10A- α cat cell migration speed was steady over time (Figure 7D) and insensitive to obstruction density.

Relative to WT cells, although the migration speed of 10A- α cat cells increased on all substrates, this rise was the most substantial in dense obstructions (Figure 7E), which eventually led to the obstruction-insensitive collective cell migration noted previously. The fluidity of 10A- α cat cell migration, assessed by plotting MSD relative to time interval (Figure 7G) and diffusivity (Figure 7H), showed reduction in sensitivity to obstruction density. Although the 10A- α cat cells were, in general, faster than WT, their directionality was reduced (Figure 7, C and F; Supplemental Figure S8), indicating the random scattering that may explain the chaotic nature of the leading edge (Figure 7A, each time frame visualizations provided in Supplemental Figure S9). This is not surprising, given the loss of cell–cell adhesions after α -catenin depletion, which is expected to abrogate both physical and biochemical signaling communication across the cell colony. Consistent with these migration results, cell-shape fluidity measured in terms of shape index was also rendered insensitive to obstruction density (Figure 7I). Overall, these results indicate that the loss of macroscale cell–cell communication via α -catenin depletion is more potent at abrogating the effects of obstructions on collective cell migration compared with the cell-level disruption via Rac inhibition. These results further indicate that matrix obstructions cause macroscale epithelial response by inducing follower-like behavior across the monolayer. When “leader phenotype” is reintroduced and cell–cell communication is disabled, obstructions do not have much of an effect on collective cell migration.

DISCUSSION

Epithelial cell migration is an important process in many biological phenomena, from cellular development to wound repair, and, its disorder can lead to diseased states, such as cancer metastasis and asthma. Because of such biological transformations, collective cell migration has been shown to adopt different modes, described in terms of jamming–unjamming and solid–fluid transitions (Treat and Sahai, 2018; Ilin *et al.*, 2020; Mitchel *et al.*, 2020). When cells migrate into external environments, they actively respond to various extracellular physical properties including confinement, stiffness, and surface geometry (Ng *et al.*, 2012; Vedula *et al.*, 2012; Nasrollahi and Pathak, 2016; Sarker *et al.*, 2019). Cells are also exposed to various topologically diverse environments such as naturally occurring physiological and mechanical heterogeneities in organs, tumors, epithelial folding, or defective basement membranes (Takeda *et al.*, 2017; Walter *et al.*, 2018; Pérez-González *et al.*, 2021). Adding an ECM perspective to this understanding of collective cell migration, our findings show that topological obstructions in the matrix can trigger physical responses spanning length scales from micrometer to millimeter in healthy epithelial cells. With microfabricated stumps on PDMS substrates for epithelial cell culture, we have shown that as the density of matrix obstructions increased, cells became slower, more disordered, and lost their directional persistence, all of which indicate a less fluidic phenotype. As a result, the leading edge of the epithelial monolayer became more disorganized when moving around dense obstructions. When the spacing between the obstructions (150 μ m) was larger than the characteristic length of obstructions (100 μ m), the overall migration was not significantly impacted. However, migration was more arrested and cell shape fluidity was reduced in dense obstructions, with smaller spacing (50 μ m). We speculate that dense obstructions cause too many breaks in the epithelial monolayer to retain the original fluidic migration phenotype.

Previous work has connected cell shape fluidity and unjammed collective cell migration to increased tractions generated by diseased cell populations (Treat *et al.*, 2009; Park *et al.*, 2015). Separately, cell and tissue stiffness have emerged as a “mechanotype”

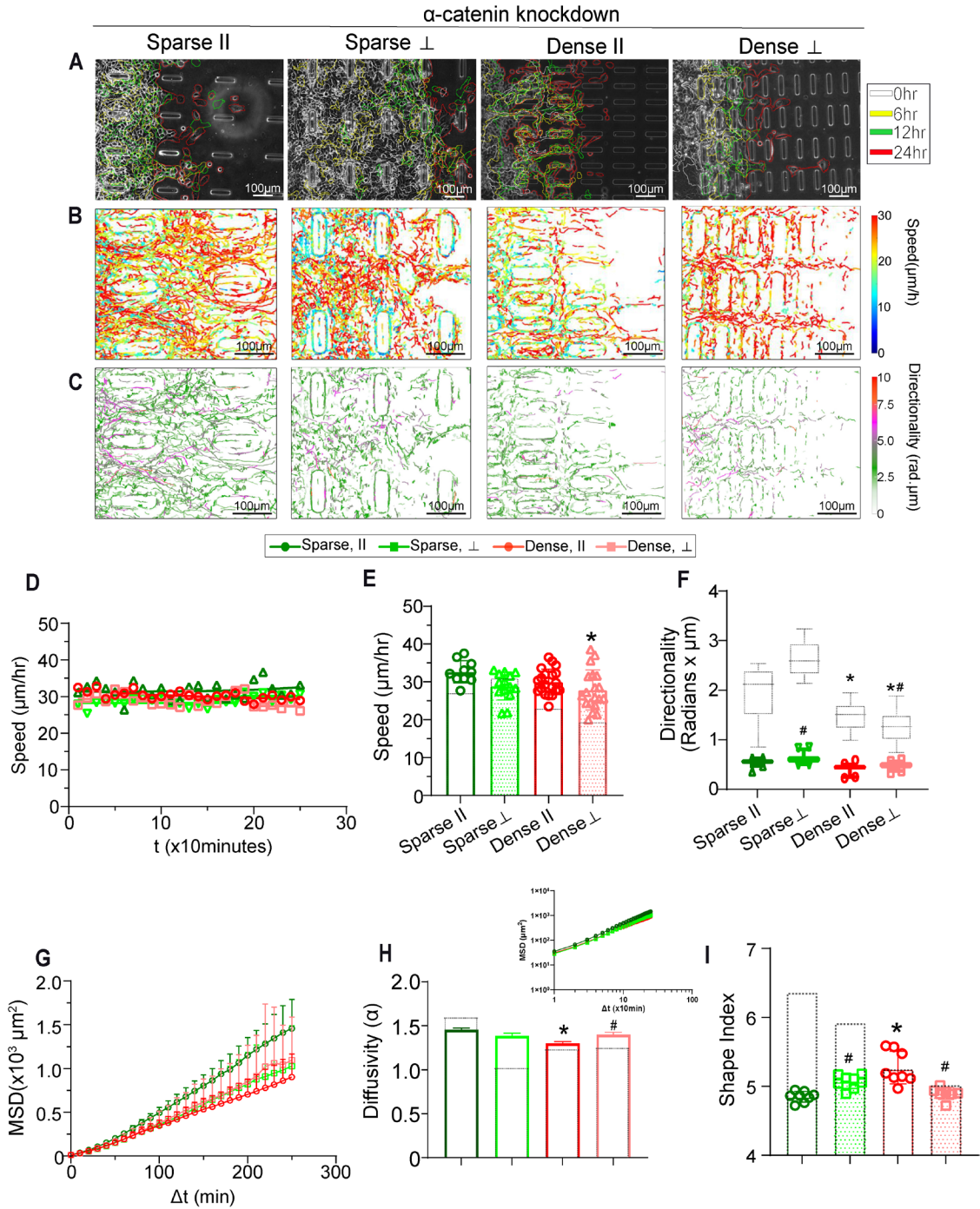


FIGURE 7: Depletion of α -catenin eliminates obstruction sensitivity in epithelial migration. (A) Representative phase contrast images at final time point (24 h after cell seeding) of MCF10A with α -catenin-KD (10A- α cat) in four substrate conditions with varying obstruction spacing and orientation. Leading edges of migrating colonies are annotated for time points at 6-h interval between 0 and 24 h. Migration trajectories of individual cells within 10A- α cat cell monolayers are plotted with color coding for time-averaged (B) speed and (C) directionality. The representative images have the migration direction of left to right. (D) Temporal variation in average migration speed for each substrate condition; points represent average across all cells for a specific time point; lines are polynomial fits across time points. (E) Migration speed of 10A- α cat cells averaged over all cells per condition; dotted lines and boxes show data for WT for direct comparisons. (F) Migration directionality of 10A- α cat cells averaged over all cells per condition. Error bars = SD. (G) Plot of mean squared displacement (MSD) versus duration of time interval Δt . (H) Cell-level diffusivity (α) calculated from the accompanying MSD- Δt log-log plot. Error bars = SEM. (I) Average cell shape index plotted for all four substrates. Error bars = SD. Significant difference annotated as * $p < 0.05$ for comparison against the sparse condition, and # $p < 0.05$ for comparison against the parallel condition. $N \geq 6$ fields of view in all plots.

for disease progression—tumor-laden tissues become stiffer (Acerbi *et al.*, 2015; Walter *et al.*, 2017; Bayer *et al.*, 2019) and cancer cells become more mechanically adaptable depending on their need for invasion (Kim *et al.*, 2016; Rianna *et al.*, 2020). Within a given epithelial population, leader cells that generally migrate faster also express higher phosphorylated myosin light chain and yes-associated protein—proteins typically associated with their mechanotransductive activation (Ng *et al.*, 2012; Sarker *et al.*, 2019). Using AFM, we performed direct measurements of cell stiffness to show that leader cells are significantly stiffer than follower cells within the monolayer. Thus, in addition to traction forces and mechanotransductive protein expression, cell stiffness gradients can also predict leader cell mechanotransduction within a given epithelial population. By performing AFM on epithelial monolayers moving through obstructions, we further show that average leader cell stiffness amongst dense obstructions reduced to the levels observed otherwise for follower cells, which indicated that obstructions could suppress physical functioning of leader cells. Further supported by reduced F-actin expression in dense obstructions, we speculate that obstructed migration may convert leader cells into follower cells in terms of their mechanical properties (Figure 8, A and B). These results show that mesoscale obstructions can cause both microscale and macroscale disruption in cell population—from the shape and stiffness of individual cells to collective migration and leading edge integrity.

Based on our AFM measurements, cell shape analyses, and computational modeling, we speculate that the phenotypic leader–follower switching during interactions with dense obstructions may give rise to the “obstruction–sensitivity” shown by healthy epithelial cells. Building on this concept, the obstruction–sensitivity could be targeted by either converting all cells into followers or leaders, thus precluding their phenotypic switching. Indeed, the suppression of the typical protrusive function of leader cells (Borgne-Rochet *et al.*, 2019; Zoeller *et al.*, 2019), performed here via Rac inhibition, reduces the migratory differences between sparse and dense obstructions (Figures 6 and 8C). Moreover, MDCK cells, a cell type with stronger epithelial characteristics and follower-like phenotype, also showed reduced sensitivity to matrix obstructions (Figures 5 and 8C). When cell communication throughout the monolayer is suppressed via α -catenin depletion in MCF10A cells, which effectively made all cells behave like individual leaders, the collective migration became more fluidic, faster, and less sensitive to obstructions (Figures 7 and 8D). When simulating migration of follower-like cells of reduced protrusions and strong adhesions through obstructions, their sensitivity to obstruction density was lowered (Figure 8E) compared with control cells. Simulations of leader-like cells with high protrusions and weak adhesions also predict lowered sensitivity to obstructions.

Taken together, both modeling predictions and analogous experimental results (Figure 8, E and F, and Supplemental Figure S8) indicate that healthy cells with optimal balance of cell–cell adhesions and protrusive behavior are more obstruction sensitive than aggressive, dysfunctional cell populations composed of either hyperactive leaders or cohesive followers. This fundamental difference in obstructions sensitivity of cohesive versus aggressive populations may also be applicable in complex collective systems other than cells. Over the past decade, the pathological transformation of cells has been associated with their ability to sense and respond to matrix stiffness (Paszek *et al.*, 2005), detach and leave populations via mesenchymal transition (Hanahan and Weinberg, 2011; Guen *et al.*, 2017), and negotiate confined environments (Beadle *et al.*, 2008; Pathak and Kumar, 2012). As such, stiffness–sensitivity, cell shape,

traction forces, confined migration, and cell stiffness have now been recognized as predictive disease markers for cells (Network *et al.*, 2013; Park *et al.*, 2015; Mitchel *et al.*, 2020), which have been used to distinguish them from healthy cell populations and perform drug screening (Lampi and Reinhart-King, 2018; Lin *et al.*, 2018). According to our findings, the obstruction–sensitivity of a given cell population may emerge as a disease marker, with lower sensitivity possibly predicting an invasive and scattered disease phenotype. In future work, our integrative methodology of obstruction-sensitive migration analyses, AFM measurements, and computational modeling presented here can be scaled up to test obstruction sensitivity of a variety of cells and situations in a high throughput manner. Because our method of PDMS-based microfabrication and the resultant substrate design (compatible with standard microscopy) can easily accommodate any number or geometries of obstructions, it can be adapted for different cell types and ex vivo tissue samples that may vary in their pertinent length scales of obstruction sensitivity.

MATERIALS AND METHODS

Photolithography for silicon masters with stumps

SU8-2050 photoresist was coated evenly on the silicon master using a spin coater (Brewer Science CEE 200x Spin Coater) and exposed to 365-nm conventional UV light. The pattern on the silicon master was in the square with the stumps implanted in the migratory direction. Then, the undeveloped photoresist was dissolved using the SU-8 developer solution (MicroChem). The processing time was determined according to the manual provided by Washington University in St. Louis clean room protocol. Then with a UV-LED mask aligner (KLOE UV-KUB3), photolithography was performed on the silicon master that has been coated. To ensure the dimension of the stumps, the measurement was performed using the KLA Tencor D-100 series profilometer.

Polydimethylsiloxane (PDMS) substrate fabrication

Silicon/SU-8 master with the stump design was treated with Trichloro (1H, 1H, 2H, 2H-perfluorooctyl) silane (Millipore Sigma, St. Louis, MO) for 30 min inside a vacuum desiccator and dried on a hotplate for 10 min at 150°C. SYLGARD-184 silicone-based elastomer polymeric base and curing agent were mixed at a 10:1 ratio, respectively, and were poured onto the mold. Then, the PDMS mixture was cured in a vacuum desiccator for 1 h at room temperature (RT), after which the temperature was increased to 70°C for 4 h. Each PDMS substrate was placed in a CL-1000 Ultraviolet Cross-linker with 254-nm bulbs for 10 min for sterilization. Surface treatment in preparation for cell culture was accomplished by surface activation in a plasma cleaner (Harrick Plasma, Ithaca, NY) for 5 min, after which the PDMS was incubated with a fibronectin solution at a concentration of 0.05 mg/ml overnight at 4°C.

Cell culture

MCF10A (ATCC) nonmalignant human breast epithelial cells and their anticatenin shRNA (α -cat KD) variant (courtesy of Greg Longmore, Washington University School of Medicine) were cultured in DMEM/F12 (Millipore Sigma) composed with 5% Horse Serum (Invitrogen), 20 ng/ml epidermal growth factor (EGF, Miltenyi Biotec), 0.5 mg/ml hydrocortisone (Sigma), 100 ng/ml Cholera Toxin (Millipore Sigma), 10 μ g/ml Insulin (Millipore Sigma), and 0.2% Normocin (Invitrogen) at 37°C with 5% CO₂. Media was changed every 3 d in culture plate. MDCK (WashU Tissue Culture Center), canine kidney epithelial cells are cultured in DMEM composed of 10% FBS (Life Technologies) and 0.2% Normocin (Invitrogen) at

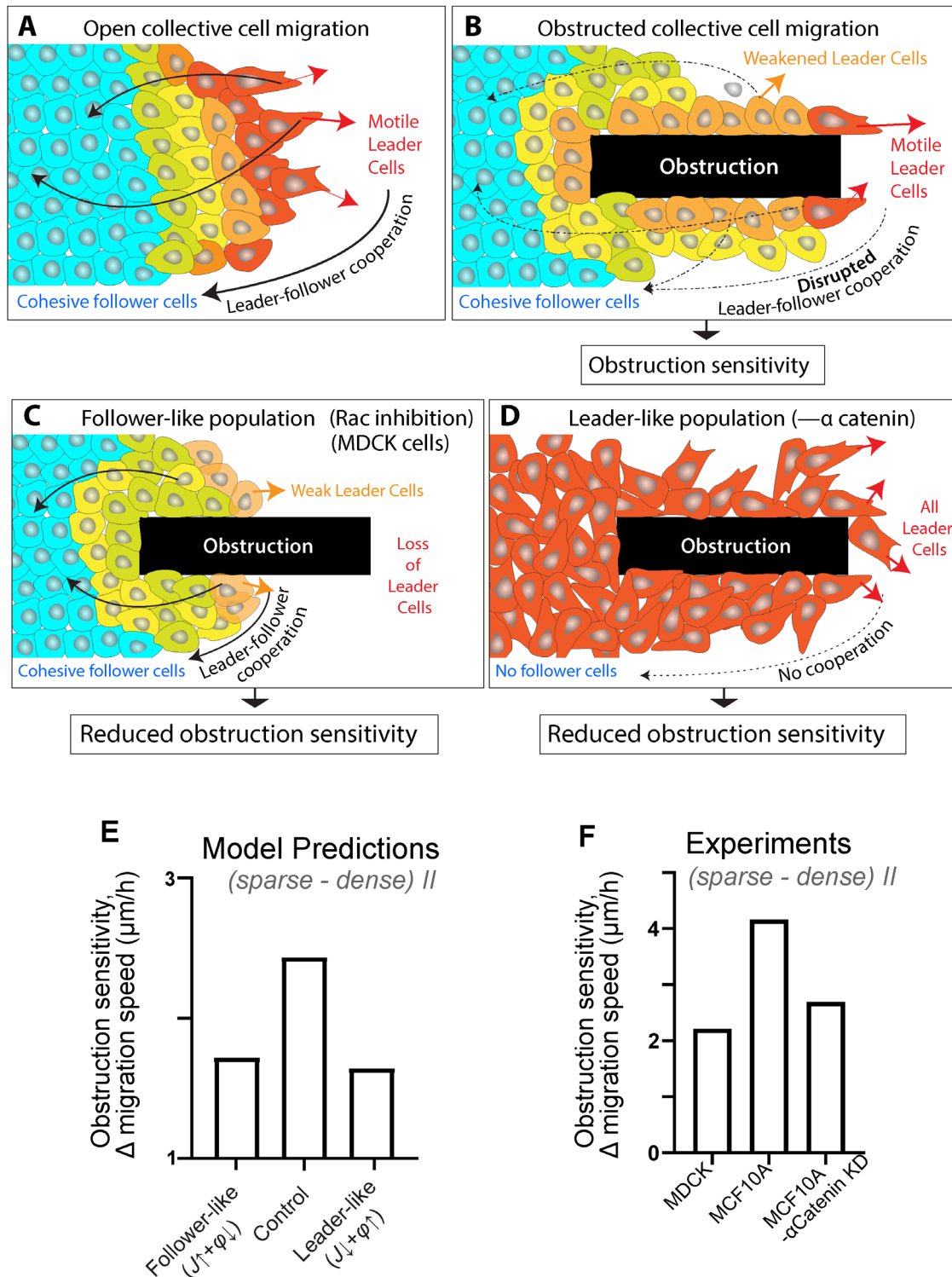


FIGURE 8: Conceptual model for obstruction sensitivity in collective migration of different cells. (A) In unobstructed “open” matrices, motile leader cells pull forward and cooperate with trailing follower cells to enable collective cell migration. (B) In the presence of dense obstructions, leader cells are weakened and leader–follower cooperation is disrupted, which slows down collective migration and generates “obstruction sensitivity.” (C) When all cells are follower-like, via Rac inhibition in MCF10A cells or naïve MDCK cells, their obstruction sensitivity is hampered because all cells in those populations are suppressed in their motility. (D) When all cells are leader-like with weak cell–cell adhesion, their obstruction sensitivity is further diminished due to a lack of macroscale communication across epithelial population. Obstruction sensitivity of migrating cells is plotted as a difference in migration speed between sparse and dense conditions from (E) modeling predictions and (F) experimental measurements corresponding to the three conditions of follower-like, control, and leader-like cell populations. Arrows indicate direction of cellular migration.

37°C with 5% CO₂. Media was changed every 3 d in culture plate. Cell line identities were authenticated using short tandem repeat (STR) profiling, and cell lines tested negative for mycoplasma contamination. To seed the cell on the microfabricated PDMS, approximately 3–5-μL (~20,000 cells) droplet was placed on the dedicated flat area (See Supplemental Figure S1A). shRNA knockdown of α-catenin was developed by inserting a lentiviral pRLRu vector (Loza et al., 2016b). Rac inhibition was performed by directly adding 20-μM NSC23766 trihydrochloride to MCF10A monolayers. The effect of NSC23766 trihydrochloride was observed to take effect within 1 h after treatment. Itraconazole (Millipore Sigma) was diluted in cell culture media and added into each well with MCF10A for duration of time-lapse imaging.

Immunofluorescence staining, imaging, and analysis

Following 3 d in culture, samples were fixed in 4% paraformaldehyde solution for 10 min or 100% methanol in 4°C for 15 min. Samples were washed and treated with 30 min of 100-nM phalloidin (Invitrogen) and 10 min of 1 μg/ml Hoechst 33342 (Thermo Fisher) in Dulbecco's phosphate-buffered saline (DPBS). For E-cadherin, 5 μg/ml anti-E-cadherin antibody [DECMA-1] intercellular junction marker (Abcam) is diluted in antibody dilution buffer and incubated in 4°C overnight. Then, secondary antibody was added and incubated for 1 h in room temperature, followed by 10 min of 1 μg/ml Hoechst 33342 (Thermo Fisher) in DPBS. Images were taken using the confocal microscope (Zeiss LSM 880; Carl Zeiss Micro Imaging, Germany) with constant acquisition parameters, using a 40× objective with a 2× digital zoom. To prevent bias across conditions, laser intensity and exposure time were kept constant, and at least two positions were taken per sample, with at least three samples per condition. With acquired images, the basal area was used to calculate the shape index of each cell through Seedwater Segmenter (Mashburn et al., 2012). The shape index was calculated according to the equation:

$$\text{Shape index (SI)} = p / \sqrt{A}$$

where p = perimeter and A = area.

Live imaging and real-time analysis

Cells (MCF10A or α-Catenin-KD, as applicable) were incubated with 1 μg/ml Hoechst 33342 for 10 min, for nuclear labeling, before running time-lapse imaging. For all conditions, images were taken with a Zeiss AxioObserver microscope (10× objective) equipped with a cell incubator chamber (Carl Zeiss Microscopy) in 10-min intervals for 3 d. The microscope stage incubator was maintained at 37°C with 5% CO₂ over this time. Images were analyzed with FIJI, the Trackmate plugin (NIH), and MATLAB particle image velocimetry. The raw data were processed to calculate directionality, MSD, speed, and correlation length according to the equations described later, which were implemented using Python, MATLAB, or R studio programming languages. Correlation length (λ) is calculated using the correlation distance function $C(r) = (1 - C_p)e^{-r/\lambda} + C_p$, where r is the distance of separation caused by migration (obtained from the particle image velocimetry data) and C_p is the function plateau, as described previously (Loza et al., 2016a; Sarker et al., 2019).

$$\text{Speed} : \sum_{i=0}^t \frac{\sqrt{((x_i - x_{i-1})^2 + (y_i - y_{i-1})^2)}}{\text{Total time}}$$

$$\text{Directionality} : = \cos^{-1} \left(\frac{v_1 v_2}{\sqrt{dx_1^2 + dy_1^2} * \sqrt{dx_2^2 + dy_2^2}} \right) * \sqrt{((x_i - x_{i-1})^2 + (y_i - y_{i-1})^2)},$$

$$\text{where } [dx_1, dy_1] = [(x_i - x_{i-1}), (y_i - y_{i-1})]$$

$$[dx_2, dy_2] = [(x_{i+1} - x_i), (y_{i+1} - y_i)]$$

Mean squared displacement (MSD): $\text{MSD}(\Delta t) = \langle |r_i(t + \Delta t) - r_i(t)|^2 \rangle$,
 $r_i(t)$ = Cell(i)'s position at the time point,
 $\langle \dots \rangle$ = average over total time.

Atomic force microscopy

AFM was performed according to previous work (Nasrollahi et al., 2017; Sarker et al., 2019). All measurements were acquired with a Bruker Nanoscope Resolve atomic force microscope (Bruker, Billerica, MA). Custom AFM probes were used, comprising a silicon nitride cantilever with a 4.5-μm diameter polystyrene bead attached at the tip (Novascan Technologies, Boone, IA), with reported stiffness of 0.01 N/m. Tips were initially calibrated in testing buffer to determine the spring constant of the cantilever. After determining the cantilever's spring constant, the probes were allowed to equilibrate in for at least 10 min to prevent drift in the measured deflection due to the testing buffer. Substrates were loaded onto the AFM stage so that the stumps on the PDMS substrate ran parallel to the AFM cantilever to reduce testing complications from the vertical stumps. Testing began where the leading edges of the monolayers were at the end of a set of stumps. Full force volume measurement runs were taken in 100-μm increments until reaching the region of the monolayer before the beginning of the stumps (see Supplemental Figure S5B for full workflow and criteria for filtering force curves). Each section was taken such that measurements were taken as close to the stumps without the cantilever touching the stumps and thereby skewing the stiffness measurements. A modified Hertz model, implemented in Nanoscope Analysis Software (Bruker, Billerica, MA), was used to analyze the force curves and export stiffness data. Sample thickness was taken into account and is reflected in the reported Young's moduli similar to previous work (Dimitriadis et al., 2002).

Statistical analysis

All box and whisker plot has a representation of mean with 90–10 percentile whiskers. Statistical significance between groups was determined using one-way analysis of variance in GraphPad Prism9, and the p values <0.05 were considered to be statistically significant, unless specified otherwise.

Lattice-based modeling of collective cell migration through obstructions

According to our experimental results, obstructions in the path of collectively migrating epithelial cells slow migration speed of individual cells by introducing mesoscale disorder and cell-level softening. Here, we develop a theoretical model and a computational framework to better elucidate these multiscale effects of obstructions on active protrusions and polarity in individual cells and overall intercellular cooperation from leaders to followers within the monolayer. Our goal is to keep the model as simple as possible to capture obstruction-sensitive collective cell migration and predict outcomes as known cellular behaviors are changed. In addition to basic energy-based rules of cellular elasticity and adhesions, we apply two phenomenological effects of obstructions based on experimental observations—upon interaction with obstructions (1) leader cells adopt a follower identity and (2) all cells reduce their active protrusions because active processes cannot go through vertical obstructions.

Cell migration as an energy minimization problem

The forward motion of a cell within a colony can be understood as dynamic minimization of energy fluctuations due to the temporal evolution of active processes from protrusions and contraction, adhesions with neighboring cells, and passive elastic properties of the cell. Cells would not move if the net change in energy is positive. We define the net change in energy (Figure 7A) as the sum of energy change required for active protrusions (ΔH_{active}), passive cytoskeletal remodeling ($\Delta H_{\text{passive}}$), and formation of cell–cell and cell–ECM adhesions ($\Delta H_{\text{adhesions}}$), written as:

$$\Delta H = \Delta H_{\text{active}} + \Delta H_{\text{passive}} + \Delta H_{\text{adhesions}} \quad (1)$$

Adhesion energy

Cellular Potts model (Graner and Glazier, 1992) represents a space as a discrete collection of lattice points—pixels on a 2D grid or voxels in a 3D grid. We model a collection of biological cells by attaching to each lattice point ($i; j$) of a square lattice a label σ_{ij} , which identifies the corresponding cell, and a label $\tau(\sigma_{ij})$, which identifies cell type. Adjacent lattice sites are defined to lie within the same cell if they have the same value of σ_{ij} . System evolves by the random movement of individual pixels that move according to transition probabilities based on Monte Carlo simulations based on the energy criterion described previously (Swat et al., 2012). At each MCS, two neighboring pixels are chosen randomly, with one as source pixel and the other as target pixel. If both pixels belong to the same cell (ie, $\sigma_i = \sigma_j$), then no changes are made to the lattice. Otherwise, the source pixel attempts to occupy the target pixel based on Monte Carlo acceptance probability, which is calculated from the difference in total system energy. The total system energy associated with the configuration, before and after the move, is defined as per Eq. 1. We provide specific definitions for each term in Eq. 1, as following:

$$H_{\text{adhesions}} = \sum_{\forall i,j; \sigma_i \neq \sigma_j} J_{\tau(\sigma(i)), \tau(\sigma(j))} \quad (2)$$

Here, $J_{\tau(\sigma(i)), \tau(\sigma(j))}$ represents contact energy for the two cell types in contact and $\tau(\sigma(i)), \tau(\sigma(j))$ represents contribution from the total energy due to cell–cell adhesions.

Passive cytoskeletal energy

Energy due to passive bulk elasticity of cells, H_{passive} , is defined as quadratic change in cell volume, as done previously (Graner and Glazier, 1992):

$$H_{\text{passive}} = \sum_{\forall \text{ cells}} K_a (V_0 - V)^2 \quad (3)$$

Here, $K_a = 5$ is a constant for bulk elasticity and $V_0 = 200 \mu\text{m}^3$ is target volume of the cells in isolation. After calculating energy of system before (H_i) and after (H_f), the copy attempt will always be successful if $H_f < H_i$, that is, $\Delta H < 0$. If $\Delta H \geq 0$, the copy attempt is accepted with a probability of $e^{-\left(\frac{\Delta H}{T}\right)}$, $T = 20 \text{ J}$. Higher values of T would tend to accept more unfavorable copy attempt.

Active protrusive energy and leader–follower cooperation

Cell migrates through coupled processes of forward propulsion via protrusions and active cytoskeletal remodeling. Adapting from the classical Glazier–Graner model (Graner and Glazier, 1992), we define net change in energy due to these active processes as:

$$\Delta H_{\text{active}} = \phi(P - P_0) - \alpha_{\text{cell}} \quad (4)$$

which combines actin cortex–mediated cytoskeletal remodeling (first term) and α_{cell} , a coarse-grained parameter for cell-specific motility, which has been used in various cell migration models (Szabo et al., 2010; Camley and Rappel, 2017). In the first term, P is current cell perimeter and $P_0 = 40 \mu\text{m}$ is the target perimeter. As such, higher cell perimeter indicates higher cell spreading, which has to be balanced by lower values of contraction parameter ϕ to enable migration. Higher values of the contraction parameter ϕ result in high energetic cost to extend protrusions. In control conditions, that is, the open stump-free surfaces, we use $\phi = 0.1$. Because cells cannot insert protrusions into obstructions, we use $\phi = 3$ when cells come in contact with obstructions to capture this loss of protrusions.

It has been shown that leader cells generate higher forces and they pull on follower cells via cell–cell adhesions (Sunyer et al., 2016; Sarker et al., 2019; Hwang et al., 2023). This leader–follower cooperation due to intercellular cooperation enables collective cell migration. Here, we define leader cells as those at the leading edge of the migrating colony. We define leader cell protrusive energy as $\alpha_L = 8$, which lowers the net energy to enable migration. For collective migration, leaders must communicate with followers, both mechanically and biochemically. We defined this intercellular communication and cooperation by defining protrusive energy of follower cells as an average of all leader cell protrusions such that follower cell

protrusive energy is $\alpha_F = \frac{\sum_{i \in N} \alpha_{L,i}}{N}$, where N is the number of leader cells. In the absence of intercellular cooperation, for example, in case of α -catenin knockdown, follower cells may also behave as leader cells and thus have the same protrusive energy. In addition, when leader cells come in contact with obstructions, we assume that they are unable to “lead” migration anymore. Thus, the protrusive energy of leader cells is reduced $\alpha_L = 8$ to $\alpha_L = 4$, when they come in contact with obstructions.

Simulation procedure

We perform cellular Potts simulations within the framework of CompuCell3D and custom Python script to model the cellular response described previously. We simulate cell colony over a defined field (700 pixels $\times$ 375 pixels, 1 px = 2 μm). Total simulation time is 4800 MCSs, equivalent to 72 h. Cell monolayer is defined as a flat collection of cells of diameter = 10 μm . Obstructions are defined as areas that cells are not permitted to occupy. Here, we assume that the simulation space is in a 3D space and show simulations focusing on the basal 2D plane of obstructions and the migrating cell colony.

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